

Progressive Ball-Carrying and Shots in the English Premier League*

Eddie Nguyen

September 25, 2026

This study examines the relationship between progressive ball-carrying and shooting activity among 279 outfield players in the 2024–25 English Premier League season. Progressive carries per 90 minutes were used as the explanatory variable and shots per 90 minutes as the response variable. A simple linear regression was used to evaluate the direction and strength of their linear association. The results indicate a statistically significant positive association between progressive carries and shots per 90 ($\hat{\beta}_1 = 0.0280$, $p = 0.0335$). However, the relationship was weak, with progressive carries explaining only 1.62% of the variation in shots per 90 ($R^2 = 0.0162$). These findings suggest that players who make more progressive carries tend to record slightly more shots, but progressive carrying alone provides limited information about individual shooting activity. The results also highlight the limitations of using a single simple linear regression to describe attacking performance, particularly given differences in player roles and the diagnostic issues observed in the model.

Introduction

The English Premier League (EPL) is one of the most competitive soccer leagues in the world, where players must perform under high levels of tactical and physical pressure. In this environment, individual ball-carrying can be an important way for players to advance the ball, bypass opponents, and move their team into attacking positions. Progressive carries are one statistic that can be used to measure a player's ability to advance the ball through individual carries. Understanding whether this type of ball progression is associated with greater attacking activity can provide insight into the role of individual ball-carrying in the EPL.

Although progressive carries measure a player's ability to move the ball forward, they do not directly indicate whether those actions contribute to attacking opportunities. A player may

*Project repository available at: <https://github.com/EddieNguyen2012/EPL-Progressive-Carries-vs-Shots-Analysis>.

frequently carry the ball forward without necessarily producing shots. Therefore, it is useful to examine whether players who make more progressive carries also tend to take more shots. This paper addresses this question by examining the relationship between progressive carries and shots among EPL players during the 2024–25 season. Since players receive different amounts of playing time, both variables are measured on a per-90-minute basis.

To investigate this relationship, I use a simple linear regression with progressive carries per 90 minutes as the explanatory variable and shots per 90 minutes as the response variable. The analysis examines the direction and strength of the linear relationship between these two measures. Our results show a **positive but weak association** between progressive carries per 90 and shots per 90. Although the relationship is statistically significant ($\hat{\beta}_1 = 0.0280$, $p = 0.0335$), progressive carries per 90 explain only **1.62% of the variation** in shots per 90 ($R^2 = 0.0162$). These findings suggest that players who make more progressive carries tend to take slightly more shots, but progressive carrying alone provides limited information about a player’s shooting activity.

The remainder of this paper is organized as follows. The Methods section describes the data, variables, and simple linear regression used in the analysis. The Results section presents the regression results and evaluates the relationship between progressive carries and shots. The Discussion section interprets the findings, considers their implications and limitations, and relates them back to the original research question. Finally, the Conclusion summarizes the main findings of the analysis.

Data

Data Trimming

The data consist of 562 players from the 2024–25 English Premier League (EPL) season, with each player representing one observational unit. The dataset was obtained from the SCORE Sports Data Repository, which compiled player-level statistics from the season 2024 - 205 (SCORE 2025).

The two variables used in the analysis are **Progressive Carries** and **Shots**. **Progressive Carries** measures a player’s carries that substantially advance the ball toward the opponent’s goal, while **Shots** records the number of shots taken by a player during the season (38 90-minutes games per season) (Opta Analyst 2024). Noted that I excluded players that played less than 30% of the total minutes of the season (due to various reasons like injuries, etc.) because their number may not reflect their true ability. Also, I excluded the Goalkeepers because their progressive carries resulting in a goal is extremely rare, almost impossible.

Table 1: Summary Statistics of Progressive Carries

Progressive.Carries	Shots
Min. : 0.0	Min. : 0.00
1st Qu.: 76.5	1st Qu.: 12.00
Median :119.0	Median : 22.00
Mean :139.2	Mean : 30.44
3rd Qu.:191.5	3rd Qu.: 43.00
Max. :494.0	Max. :130.00

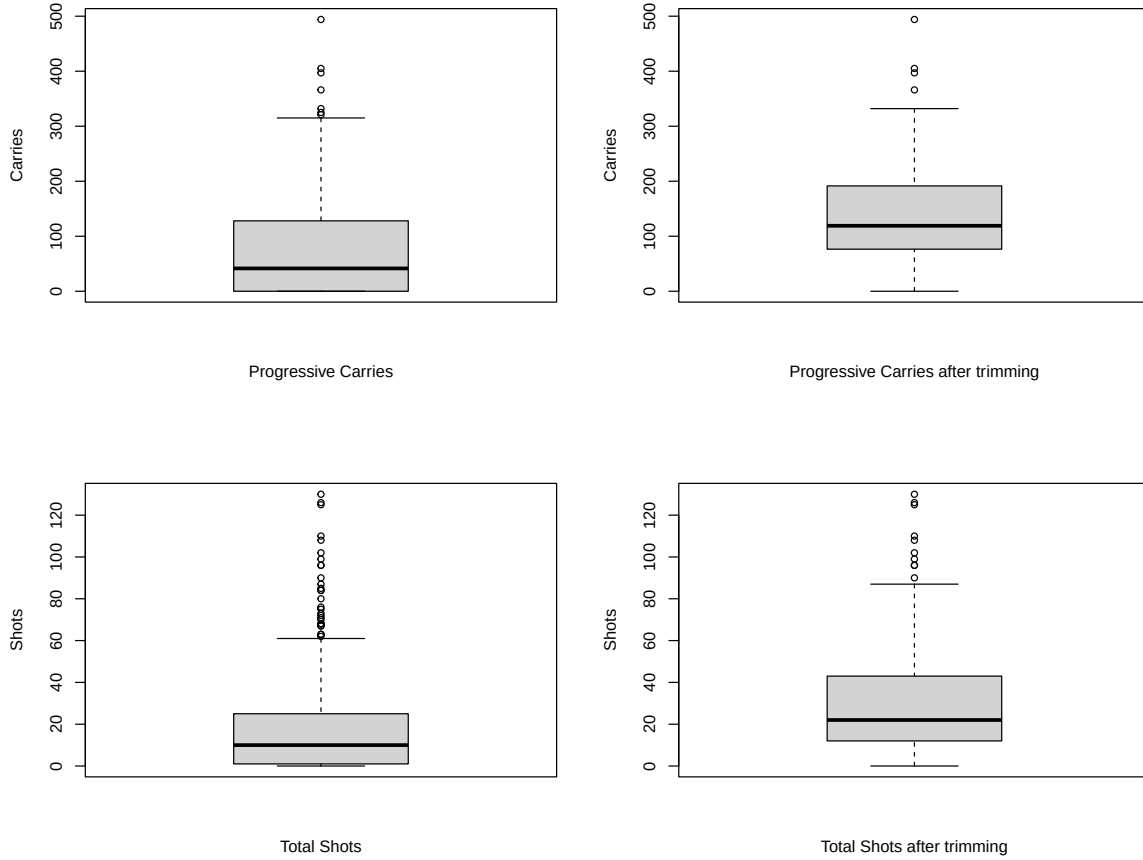


Figure 1: Comparison of Progressive Carries and Shots before and after filtering

Figure 1 displays the boxplots of both metrics after excluding goalkeepers and applying the 70% total playing time filter and only outfield players filters ($n = 279$). The distribution of

Total Shots per 90 is well-behaved and largely symmetric, with a median of approximately 1.0 shot per 90 and a single legitimate high-volume outlier near 4.0. Progressive Carries per 90 exhibits moderate right-skewness, showing a small subset of specialist ball-carrying wingers who average between 15 and 28 progressive carries per 90.

Scaling To Per 90

Because players have different amounts of playing time, I also used **Minutes**, which recorded the exact amount of time that player featured on the field throughout the whole season. **Progressive Carries** and **Shots** variables were converted to rates per 90 minutes (a match length) as follows:

$$\text{Progressive Carries per 90} = \frac{\text{Progressive Carries}}{\text{Minutes}} \times 90$$

$$\text{Shots per 90} = \frac{\text{Shots}}{\text{Minutes}} \times 90.$$

Progressive Carries per 90 is used as the explanatory variable and **Shots per 90** as the response variable. The original dataset reports no missing values. However, players with very few minutes may have unusually large per-90 rates, so the distribution of playing time will be examined before fitting the model.

The analysis uses the original SCORE dataset as its primary data source. The main data transformation consists of converting the two season totals into per-90 rates. Summary statistics and distributions of the two variables will be examined using tables and plots, followed by a scatterplot of **Progressive Carries per 90** against **Shots per 90** to assess the form and strength of their relationship.

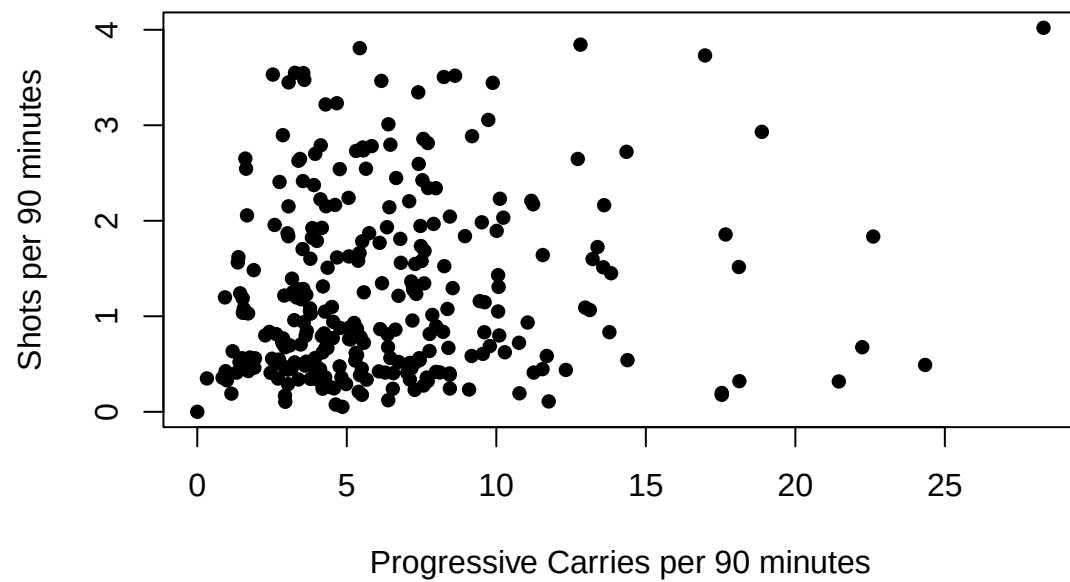


Figure 2: Progressive Carries per 90 vs Shots per 90

Figure 2 suggests a positive association between the two variables, although there is substantial variability in **Shots per 90** across players with similar rates of progressive carries. A small number of observations have substantially higher **Progressive Carries per 90** than most players, which will be considered when assessing the fitted regression model.

Separating by Position

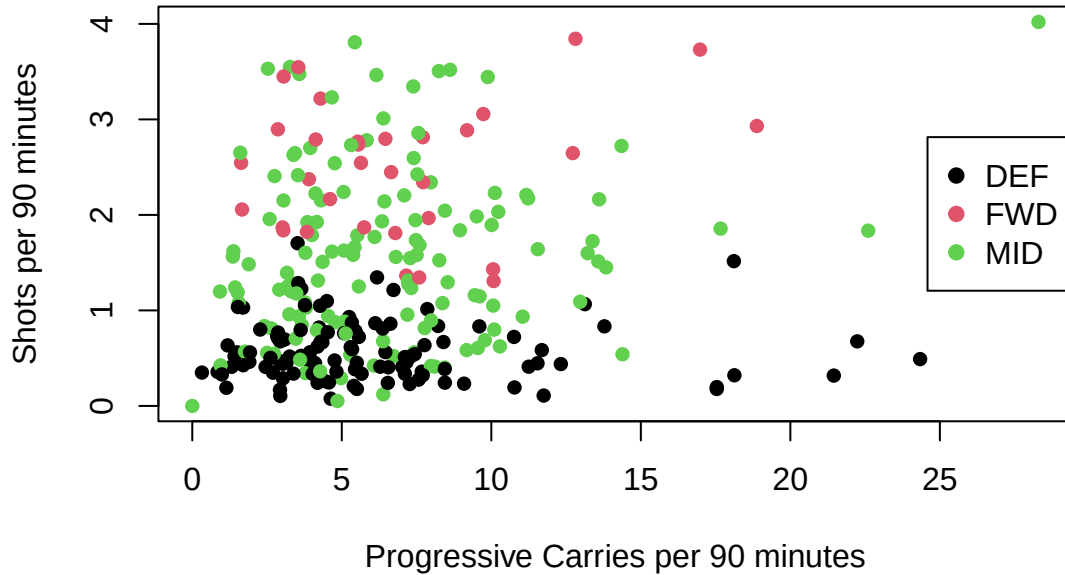


Figure 3: Progressive Carries per 90 vs Shots per 90 by Position

When separating the relationship by `Position`, I can clearly see that progressive carries from forwards and midfielders are more likely to result in higher than 1 shot per match than defenders. This reflects the nature of the roles where defenders are less likely involved in attacking sequences and their progressive carries are mostly to escape pressure from the opponents and progress the ball to players higher up the field.

Methods

Computational Implementation

All data manipulation, model estimation, and diagnostic evaluations were executed in R (R Core Team 2025) using the `tidyverse` (Wickham et al. 2019) and `dplyr` (Wickham et al. 2023) software packages.

Model Definition & Specification

I estimate a simple linear regression model (Kutner et al. 2005) to examine the linear relationship between progressive ball-carrying and shooting output among outfield players in the 2024–25 English Premier League season. Ordinary Least Squares (OLS) is used to estimate the model parameters.

The statistical model is specified as:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i \quad \text{for } i = 1, 2, \dots, n$$

where for each player i : - Y_i is the continuous outcome variable representing **Shots per 90 minutes** (Shotsper90).

- X_i is the continuous explanatory variable representing **Progressive Carries per 90 minutes** (PrgCarriesper90).
- β_0 is the intercept parameter, representing the baseline expected Shots/90 when a player averages zero progressive carries per 90 minutes.
- β_1 is the slope parameter (the primary parameter of interest), representing the expected change in Shots/90 associated with a one-unit increase in PrgC/90.
- ϵ_i is the unobserved random error term for player i , representing individual variation in shooting output unexplained by progressive carrying.
- n is the sample size of qualified outfield players ($n = 279$).

Model Selection & Validation

To minimize distortion from small sample sizes and unstable rate metrics, players were filtered using a minimum playing time threshold (e.g., at least 450 minutes played in the season). Model selection focused on isolating the bivariate association between carrying progression and shot generation. Overall model fit and explanatory power are validated using the coefficient of determination (R^2) to evaluate the proportion of outcome variance explained by the model, alongside an F -test to confirm overall statistical significance.

Model Assumptions & Diagnostics

Per the Gauss-Markov Theorem (Kutner et al. 2005) and standard regression diagnostic frameworks, OLS yields the Best Linear Unbiased Estimator (BLUE) under specific error conditions, I evaluated the six core regression assumptions using domain context and visual diagnostic plots by the decreasing structural importance,:

1. **Representativeness (Met):** The dataset includes all qualifying Premier League outfielders in the 2024–25 season, making conclusions generalizable to different EPL post-season analysis.
2. **Additivity and Linearity ($E[\epsilon_i | X_i] = 0$):** The relationship between X and Y is additive and linear, meaning the expected value of errors conditional on X_i is zero.
3. **Uncorrelated Errors ($\text{Cov}(\epsilon_i, \epsilon_j) = 0$ for $i \neq j$) (Met):** It is reasonable to assume that observations represent season-level player aggregations. However, even though players are modeled independently, mild tactical dependencies exist among players from the same team.
4. **Homoscedasticity ($\text{Var}(\epsilon_i | X_i) = \sigma^2$):** Error variance must remain constant across all levels of predicted values.
5. **Normality of Errors ($\epsilon_i \sim N(0, \sigma^2)$):** Residuals should follow a normal distribution for exact finite-sample inference.

Analytical Limitations

Several limitations should be considered when interpreting these results:

- **Omitted Variable Bias:** The simple linear regression model omits key confounding factors such as team possession dominance, player position (e.g., wingers vs. central defenders), and shot creation originating from passing rather than carrying.
- **Non-Causal Interpretation:** The model identifies a predictive linear association, not a causal mechanism; increasing a player’s carry volume will not automatically increase shot generation without contextual opportunities.
- **Spatial/Team Clustering:** Players belonging to the same club share strategic playstyles and teammates, violating strict independence across individual error terms.

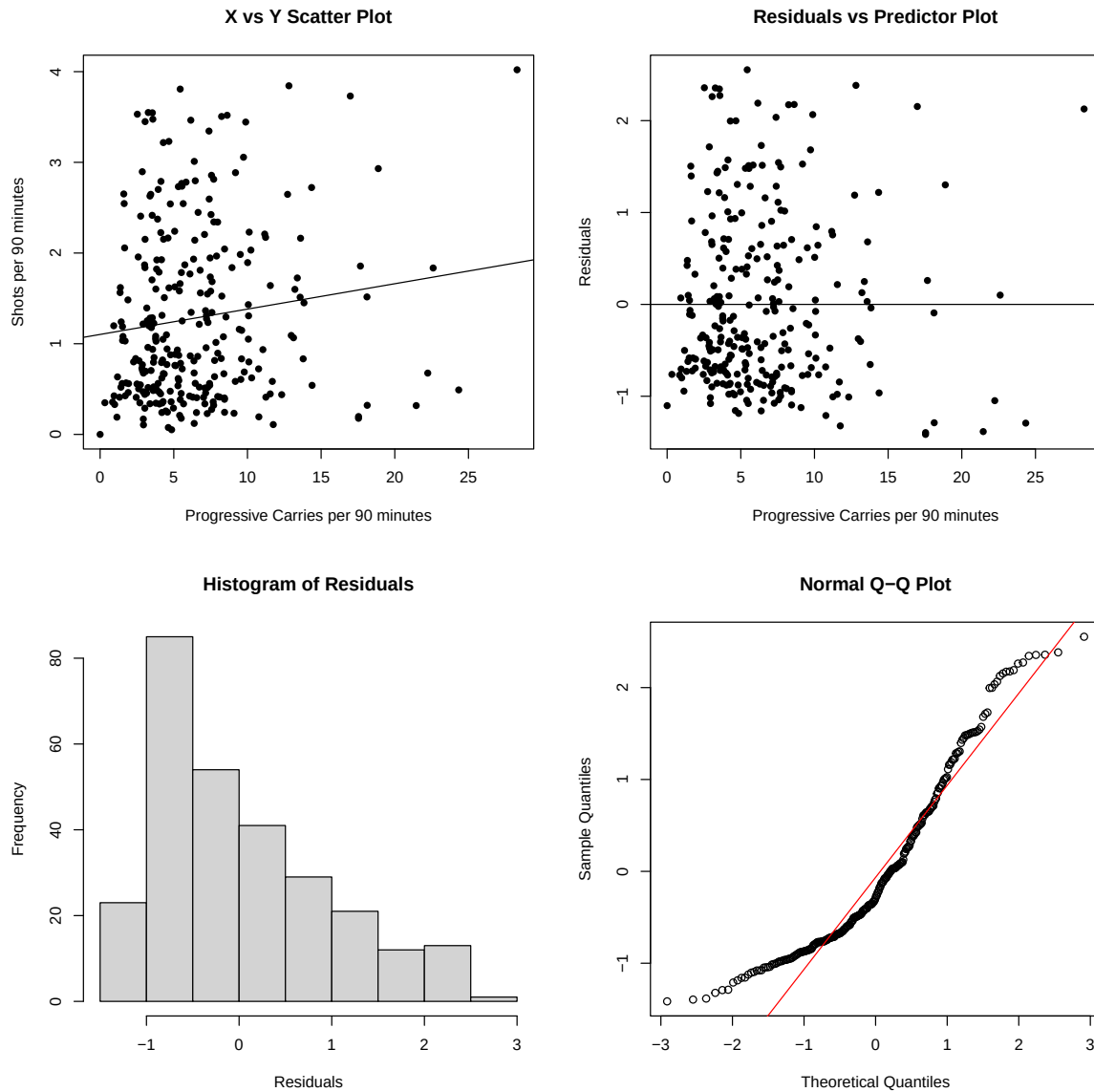


Figure 4: Diagnosis plots for the model

Evaluation:

I evaluate the core error assumptions using visual inspection of the four diagnostic plots:

1. **Linearity (Met):** Evaluated via the *X vs Y Scatter Plot* and *Residuals vs Predictor Plot*. The scatter plot reveals a positive linear relationship between progressive carries

and shots per 90. In the residuals plot, errors are centered around the horizontal zero line across the range of X_i , confirming that the zero conditional mean assumption holds, despite data sparsity at high carrying volumes ($X_i > 15$).

2. **Homoscedasticity (Met):** Evaluated via the *Residuals vs Predictor Plot*. The error variance exhibits mild heteroscedasticity. Because shooting output cannot be negative, a hard lower boundary exists for residuals (bounded near -1.0 to -1.4), whereas positive errors extend beyond $+2.5$. This creates an asymmetric vertical spread across predictor levels.
3. **Normality of Errors (Met under CLT):** Evaluated via the *Histogram of Residuals* and *Normal Q-Q Plot*. Both plots indicate moderate positive (right) skewness. The histogram displays a prominent left-leaning peak with a tail extending to $+3.0$, while the Q-Q plot shows sample quantiles curving above the red theoretical line in both upper and lower tails. While exact finite-sample normality is violated, the sample size ($n = 279$) is sufficiently large for the Central Limit Theorem (CLT) to guarantee valid asymptotic hypothesis testing (t -tests, confidence intervals, and p -values).

So in general, all of the assumptions are satisfied, which helps us guarantee that I can draw reliable conclusions from the model.

Results

Using Ordinary Least Squares (OLS) estimation on the sample of $n = 279$ qualifying Premier League outfielders, I evaluate the linear relationship between progressive ball-carrying and shooting volume. The fitted regression model is specified as:

$$\widehat{\text{Shotsper90}}_i = 1.1022 + 0.0280 \times \text{PrgCarriesper90}_i$$

Table 2 shows the details of the estimated regression coefficients, standard errors, t -statistics, and corresponding p -values.

Call:

```
lm(formula = Shots.Per90 ~ Progressive.Carries.Per90, data = filtered_df)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.4152	-0.7407	-0.2872	0.6101	2.5535

Coefficients:

Estimate	Std. Error	t value	Pr(> t)
----------	------------	---------	----------

(Intercept)	1.10220	0.10099	10.914	<2e-16 ***
Progressive.Carries.Per90	0.02797	0.01309	2.137	0.0335 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9397 on 277 degrees of freedom

Multiple R-squared: 0.01622, Adjusted R-squared: 0.01267

F-statistic: 4.567 on 1 and 277 DF, p-value: 0.03348

Table 2: OLS Regression Results for Shots per 90 Minutes

The regression results show a positive relationship between progressive carries per 90 minutes and shots per 90 minutes. The coefficient for progressive carries is 0.02797, meaning that for every additional progressive carry per 90 minutes, the expected number of shots increases by about \$0.028 per 90 minutes. The p-value is 0.0335, which is below 0.05, so the relationship is statistically significant.

However, the model explains very little of the variation in shots (Figure 5). The R^2 value is only 0.0162, meaning that progressive carries explain about 1.62 of the differences in shots per 90 minutes. The adjusted R^2 is similar at 0.0127. This suggests that although there is a statistically significant relationship, progressive carries alone are not a strong predictor of how often a player shoots.

The residual standard error is 0.9397, meaning that the model's predictions are typically off by about 0.94 shots per 90 minutes. The overall F – test is also statistically significant ($F(1,277) = 4.567$, $p = 0.0335$), confirming that the model has a significant relationship with shots per 90 minutes. Overall, the results suggest that progressive carries are related to shooting, but the relationship is relatively small and most of the variation in shooting comes from other factors.

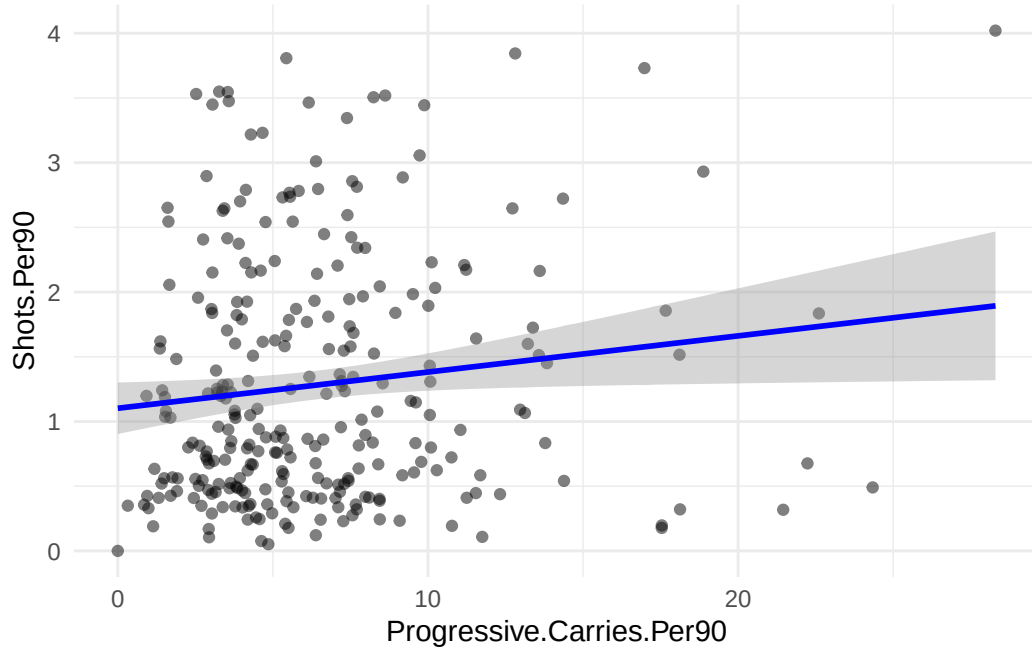


Figure 5: Fitted OLS Regression Line overlaying Shots per 90 vs. Progressive Carries per 90.

Discussions

Key Finding

Among 279 EPL regular outfield players, progressive carries per 90 were positively associated with shots per 90 ($\hat{\beta}_1 = 0.0280$, $p = 0.0335$). However, the relationship was weak, with progressive carries explaining only 1.62% of the variation in shots per 90 ($R^2 = 0.0162$). Thus, while the association is statistically significant, progressive carrying alone provides limited information about a player's shooting activity.

The results should be interpreted with several limitations. The simple regression does not account for factors such as player position, team tactics, or possession, which may contribute substantially to shooting activity. In addition, the residual diagnostics indicate that the OLS assumptions are not fully satisfied, particularly regarding variance and the distribution of the residuals. Players from the same team may also share tactical environments, which is not captured by the model.

Overall, the findings suggest that progressive carrying and shooting are related, but progressive carries alone are not a strong predictor of shots per 90. Future analyses could account for positional differences and team-level effects and consider models better suited to non-negative, skewed player statistics.

Potential different relationship by Position

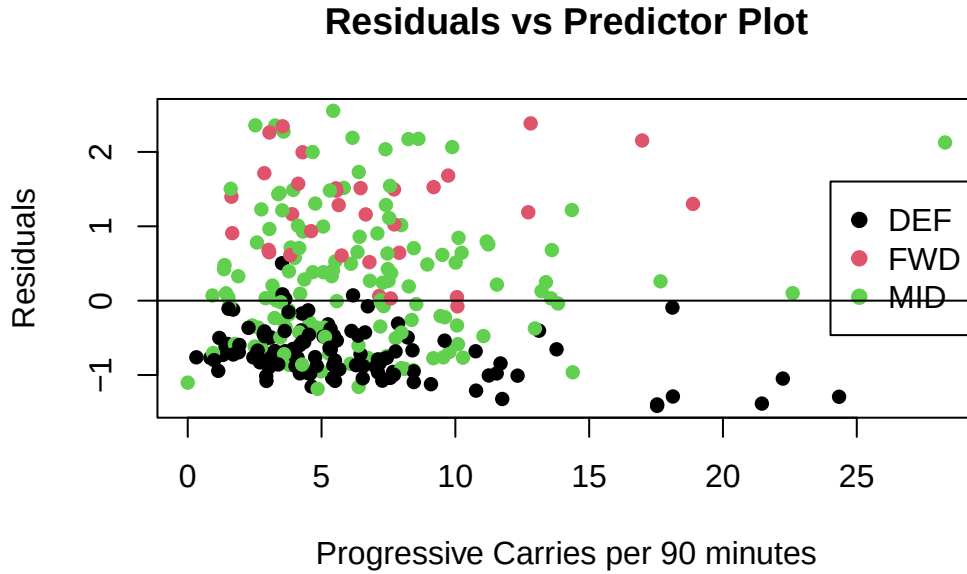


Figure 6: Residuals vs Progressive Carries per 90 by Position

As shown in the Data section, if we separate the residuals by Position, the relationship may differ between the Midfielders + Forwards versus Defender by the way each role functions naturally. This indicates that position may need to be accounted for, either by fitting separate models for each position or by including position and position–predictor interactions in a combined model.

AI Use Statement

ChatGPT (OpenAI) was used for language polishing, rephrasing, and assistance with R Markdown/Quarto syntax. All statistical analyses, interpretations, and conclusions were independently reviewed by the authors.(OpenAI 2026)

References

Kutner, Michael H., Christopher J. Nachtsheim, John Neter, and William Li. 2005. *Applied Linear Statistical Models*. 5th ed. McGraw-Hill/Irwin.

- OpenAI. 2026. *ChatGPT*. <https://chatgpt.com/>.
- Opta Analyst. 2024. *Opta Football Stats Definitions*. <https://theanalyst.com/articles/opta-football-stats-definitions>.
- R Core Team. 2025. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>.
- SCORE. 2025. *English Premier League Player Stats 2024–25*. SCORE Sports Data Repository. <https://data.scorenetwork.org/soccer/epl-player-stats-2425.html>.
- Wickham, Hadley, Mara Averick, Jennifer Bryan, et al. 2019. “Welcome to the tidyverse.” *Journal of Open Source Software* 4 (43): 1686. <https://doi.org/10.21105/joss.01686>.
- Wickham, Hadley, Romain François, Lionel Henry, Kirill Müller, and Davis Vaughan. 2023. *Dplyr: A Grammar of Data Manipulation*. <https://CRAN.R-project.org/package=dplyr>.